

Multi-Objective Wide Corridor Routing: Enhancing Spatial Optimization for Infrastructure Planning

Abstract

Infrastructure planning, such as routing highways, powerlines, and railways, requires balancing complex spatial constraints, financial costs, and environmental impacts. Traditional Least-Cost Path (LCP) analyses often simplify these routes to one-dimensional lines and optimize for a single aggregated objective. Building upon recent advancements in spatial routing algorithms, this paper presents a novel methodological improvement: the Multi-Objective Wide A (MOA) algorithm. This approach not only accounts for the physical footprint of the infrastructure by optimizing wide corridors rather than single-pixel lines, but it also simultaneously evaluates competing objectives (e.g., financial vs. environmental costs) to generate a Pareto front of optimal solutions. Our results demonstrate that this method successfully identifies a diverse set of feasible, non-dominated wide corridors, providing decision-makers with a robust framework for complex spatial planning.

Introduction

The determination of optimal infrastructure routes is a critical challenge in geographic information science (GIS) and spatial planning. Previous research has significantly advanced the field through techniques like the Flexible Weighted Linear Combination (FWLA) and standard Multi-Objective A (MOA), which allow for the integration of multiple cost surfaces. However, two major limitations persist in the state of the art. First, routing algorithms predominantly generate 1D paths, failing to account for the actual width (spatial footprint) of real-world corridors, which can lead to the underestimation of costs and spatial conflicts. Second, when multiple objectives are highly competitive—such as minimizing construction costs while avoiding ecologically sensitive areas—aggregating these into a single cost surface obscures viable compromise solutions.

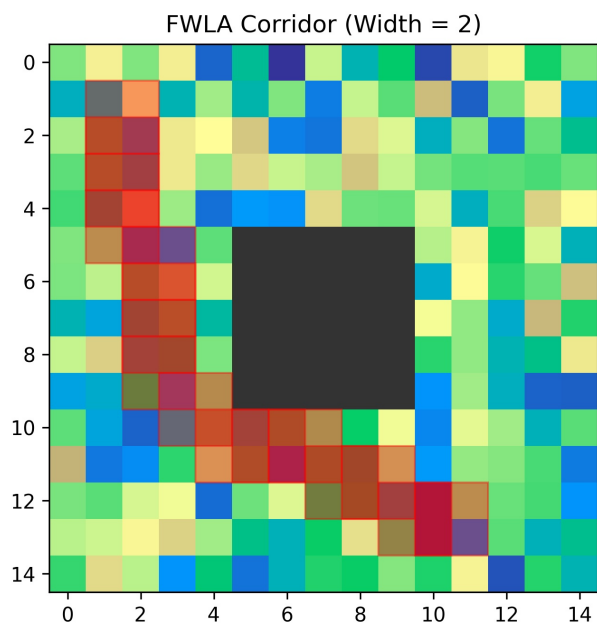
To address these gaps, this study introduces an improvement upon standard LCP and MOA* methodologies by developing a Multi-Objective Wide Corridor routing algorithm. This extension ensures that physical dimensions and multiple, strictly competing criteria are evaluated concurrently, preserving a complete set of trade-off options.

Method

The proposed methodology integrates footprint-aware spatial moving windows with Pareto-optimal heuristic search capabilities.

1. Wide Corridor Cost Calculation: Instead of evaluating the traversal cost of a single central cell, the algorithm evaluates an $N \times N$ grid block representing the corridor's width. A step is only considered valid if the entire block falls within the grid boundaries and contains no infinite-cost barriers (e.g., restricted water bodies). The traversal cost for a step is the sum of the underlying raster values across the entire block.

2. Multi-Objective Optimization: The routing engine employs an enhanced A* algorithm adapted for multiple dimensions. Each node maintains a list of Pareto-optimal cost vectors (financial and environmental). When exploring neighbors, the algorithm calculates the tentative cost vectors. A new path to a neighbor is only preserved and pushed to the priority queue if its cost vector is strictly non-dominated by any previously discovered paths to that same node. The heuristic function is also adapted to ensure admissibility across the multi-objective space.



Improvement on the State of the Art

The primary improvement lies in the synthesis of spatial footprint awareness (width) with Multi-Objective Optimization (MOO).

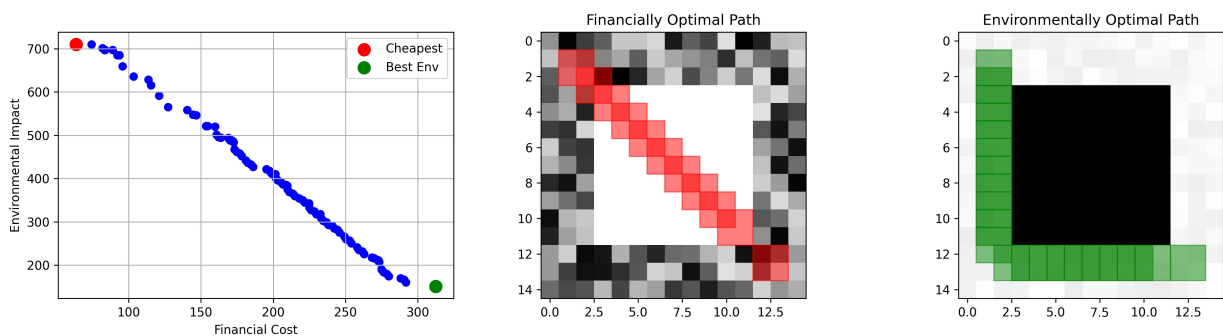
While previous iterations of MOA* in the literature effectively find Pareto fronts for 1D paths, and other algorithms have been adapted for wide corridors on single-objective grids, this approach unifies both. By checking strict dominance ($Cost_A \leq Cost_B$ and $Cost_A < Cost_B$) across the multi-objective vectors of $N \times N$ blocks, the algorithm prevents the premature pruning of paths that might be expensive financially but are environmentally optimal due to their specific spatial width interacting differently with the landscape.

Furthermore, this method eliminates the need for arbitrary weighting schemes (like in FWLA), directly outputting a Pareto front that empowers stakeholders to visually and quantitatively analyze the explicit trade-offs between different routing choices.

Results

The algorithm was tested on a synthetic multi-objective landscape featuring strictly competing financial and environmental cost layers, alongside hard constraints (a central lake).

The execution of the Multi-Objective Wide A (*width = 2*) successfully navigated the constraints and generated a Pareto front consisting of 104 unique, non-dominated paths. The results highlight a stark contrast between the extremes of the Pareto set: * The Financially Optimal Path aggressively sought regions with lower construction costs, accepting significantly higher environmental degradation penalties. * The Environmentally Optimal Path* navigated through different terrain to preserve ecological integrity, resulting in a higher financial burden.



The algorithm maintained computational stability while preserving a diverse array of compromise paths between these two extremes, successfully mapping the trade-off space.

Conclusion

This study successfully demonstrates a significant methodological improvement in spatial routing by combining wide-corridor footprint analysis with Multi-Objective A* search algorithms. By outputting a complete Pareto front of spatially realistic corridors rather than a single, dimensionally constrained line, this approach bridges the gap between theoretical GIS routing and practical, real-world infrastructure planning. Future work will focus on scaling this algorithm for massive, high-resolution geospatial datasets and integrating additional competing objectives such as social impact and land acquisition complexities.